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Dear hiring manager,

I am writing to express my interest in the Tech Lead - Advanced Analytics position. With a Ph.D. in Physics, over 10 years of experience in machine learning and python and 6 years of experience leading data science projects, and as a former intern at Total Energies Geneva, I am excited at the prospect of returning to a company at which I truly enjoyed working.

In my current role as Lead AI Architect at BMW Group, I lead the development of the onboard machine learning platform that enables efficient scheduling, monitoring, and execution of AI workloads on resource-constrained automotive hardware. I act as a technical lead in my team's research and evaluation of emerging AI technologies, with a particular focus on deploying small language models (SLMs) on neural processing units (NPUs) to enable real-time, on-device inference. Additionally, I lead our current efforts to enable, optimize and deploy SLMs for the next-generate in-vehicle voice assistant, contributing both as product owner and developer to our production-grade codebase.

Previously, at CARIAD SE, a Volkswagen subsidiary, I spearheaded the development of in-vehicle data selection triggers to solve the "long-tail problem" of autonomous driving, which were successfully deployed in the VW Group's Pioneering Fleet for autonomous vehicle training data collection. As a founding member of the department, I established and led a team of 15 software engineers, data scientists, and computer vision experts. My responsibilities also extended to active participation in the research and prototyping of deep learning-based corner case detectors, as well as constructing pipelines for active learning and benchmarking experiments. Leveraging my knowledge of statistics, I constructed statistical models for estimating the empirical test effort for autonomous vehicles. The in-depth analysis was used to guide executive decision-making.

Prior to my current role, I worked at ExxonMobil's Upstream Integrated Solutions Company in Spring, Texas, where I used a combination of deep learning, time series analysis, classical machine learning, and my background in physics to solve complex business problems. At ExxonMobil and CARIAD, I had the opportunity to mentor and supervise graduate student interns, actively contributing to their thesis work.

My technical skills encompass expertise in generative AI, agentic workflows, computer vision, statistics, edge AI, and proficiency in Python and other modeling tools. I am also experienced with research, experimentation and causal observational analysis.

Thank you for considering my application. Please find my resume attached for your reference, and I am available at your convenience for an interview.

Sincerely,

Liang Yu